

MGBOH EJIOFOR FREDRICK

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PROFESSIONAL SUMMARY

Electronics Engineering graduate with expertise in electrical systems, power generation, telecommunications infrastructure, and industrial operations. Proven track record in system maintenance, technical deployment, and equipment troubleshooting within high pressure operational environments. Proficient in Python, SQL, and Microsoft Excel for data driven system optimization and operational reporting. Demonstrated success improving system reliability, reducing downtime, and supporting large scale technical operations. Committed to safety compliance, operational excellence, and sustainable engineering solutions.

SKILLS

Programming and Data: Python (Pandas, NumPy), SQL, Microsoft Excel (Power Query, Pivot Tables), Data Cleaning, Data Validation, Data Entry, Git, GitHub

Machine Learning: Machine Learning, Hyperparameter Tuning, Model Calibration, ROC-AUC, Classification Models, Cross-Validation, Feature Engineering, Predictive Modeling

Engineering and Systems: Electrical Systems, Electronic Systems, Power Generation, Energy Systems, Telecommunications, Network Troubleshooting, System Troubleshooting, Hardware Support, Software Support, Embedded Systems, Sustainability Concepts Project and Operations: System Optimization, Process Improvement, Technical Documentation, Cross Functional Collaboration, Stakeholder Engagement, Operational Reporting.

WORK EXPERIENCE

Telecommunication Engineer | Nigerian Television Authority | DEC 2024 - NOV 2025

Broadcast system reliability as measured by sustained reduction in network downtime incidents by performing scheduled preventive maintenance, proactively troubleshooting telecommunications infrastructure faults, and applying continuous improvement protocols to technical operations. Ensured continuous live broadcasting operations as measured by uninterrupted transmission hours during critical coverage periods by diagnosing and resolving technical faults within strict timeframes, coordinating with engineering teams during infrastructure deployments, and maintaining technical documentation for equipment and network protocols.

ICT Technician | National Electoral Commission | APR 2023

Improved election technology readiness as measured successful hardware and software deployments across multiple polling centers by installing and configuring systems, performing system validation checks, and creating technical documentation for operational procedures and troubleshooting guides. Minimized onsite technical disruptions during election operations as measured by rapid resolution of all reported incidents by providing hardware and software support to polling staff, implementing continuous improvement feedback loops based on field issues, and managing stakeholder communications between technical teams and election officials.

PROJECTS

RESEARCH AND ENGINEERING PROJECTS

Smart Touchless Waste Management System: Designed and developed a sensor-based automated waste system enabling contactless disposal using embedded system principles Integrated control logic and real-time response mechanisms to improve hygiene and operational efficiency Applied low-power design considerations, aligning with energy-efficient system development

Demonstrates application of engineering solutions to sustainability, smart environments, and urban waste challenges

- Designed and developed a sensor based automated waste disposal system using embedded system principles and contactless disposal mechanisms
- Integrated control logic and real time response mechanisms to improve hygiene and operational efficiency in urban environments
- Applied low power design considerations to align with energy efficient system development and sustainability goals

DATA ANALYSIS AND SYSTEM OPTIMIZATION | PERSONAL PROJECT

- Automated Excel cleaning workflows to reduce manual processing time and improve data accuracy
- Removed duplicates, corrected formats, and validated thousands of records to prepare analysis ready datasets
- Created datasets suitable for reporting and machine learning tasks through systematic data preprocessing

MACHINE LEARNING AND PREDICTIVE MODELING

Classification model reliability as measured by optimized ROC-AUC performance metrics by conducting systematic hyperparameter tuning across candidate algorithms, applying probability calibration techniques to reduce prediction bias, and producing technical documentation for reproducible machine learning workflows.

Predictive accuracy consistency as measured by calibrated probability distributions aligning closely with actual outcomes by implementing cross validated hyperparameter optimization strategies, evaluating model calibration curves against baseline performance, and refining feature engineering pipelines to support data driven decision making.

EDUCATION AND CERTIFICATIONS

BACHELOR'S DEGREE | Electronics Engineer | University of Nigeria, Nsukka
CERTIFICATION | Certificate of Presentation | Nigerian Institute of Electrical and Electronics Engineers

SEPT 2017 – JULY 2023
MAY 2023

VOLUNTEER EXPERIENCE

Web developer | WORFAF (Women and Orphans Rights to Family Assets Foundation)

DEC 2024

LANGUAGES

English – Fluent
German – Basic
Igbo - Native